

# Ahmed Fathy

Software Engineer

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## EDUCATION

- **Cairo University, Faculty of Engineering** Giza, Egypt  
*B.Sc. in Computer Engineering; GPA: 3.7* 2022 – 2027

## EXPERIENCE

- **Al Nada Scientific Office** | *Software Developer (Part-time)* Cairo, Egypt | June 2022 – 2025
  - **ERP System:** Developed a local desktop application (packaged with Electron) managing customers, suppliers, inventory, and sales records using React.js, Express.js, and PostgreSQL.
  - **Corporate Web Solution:** Built the official company website integrated with an e-commerce platform designed to seamlessly manage and showcase sponsors, partners, services, and dynamic product catalogs.

## PROJECTS

- **TCCD Official Website & Management System** | *Frontend Development — React, Typescript* 2026
  - **Impact:** Developed a comprehensive student activity platform that successfully scaled to support and manage over 20,000 active users.
  - **Core Features:** Engineered a robust event and judging system that handles registration slots, waiting lists, and distinct role-based access for members, standard users, and corporate entities.
  - **Administration:** Built an advanced admin dashboard equipped with user-facing analytics to streamline the management of events, user data, and company profiles.
- **Tactic (Robotics & AI)** | *Embedded Systems* 2024
  - **System Integration:** Developed browser-controlled robotics with AI vision and low-latency control
  - **Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control.
  - **Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.
- **Image Restoration Pipeline** | *Deep Learning & Computer Vision, Python* 2024
  - **Pipeline:** Constructed end-to-end restoration pipeline: Denoising, Deblurring, and Inpainting.
  - **Algorithms:** Applied Telea (FMM) inpainting and deep learning models to restore damaged historical photographs.
- **Machine Fault Listener System** | *Machine Learning & Audio Processing* 2026
  - **Pipeline & Architecture:** Designed and implemented an end-to-end machine learning pipeline to classify industrial audio signals into 6 distinct operational states (normal/abnormal).
  - **Optimization:** Balanced model accuracy with inference speed under strict computational constraints (maximum 2B parameters), utilizing deep learning techniques for feature extraction and classification.
- **Autonomous Wall-Following Robot** | *Embedded Systems & C/C++* 2026
  - **System Architecture:** Designed a 5-state FSM to manage autonomous navigation, turning, and data flow.
  - **Low-Level Firmware:** Programmed STM32/AVR microcontrollers at the register level to optimize hardware timer usage.
  - **Control & Integration:** Combined proportional control, ultrasonic/gyro sensors, and WiFi for real-time tracking and telemetry.

## STUDENT ACTIVITIES

- **IEEE** | *UI/UX Designer - Frontend Team Leader* Cairo | Oct 2025 – Present
  - **Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **Technical Center for Career Development (TCCD)** | *Frontend Developer* Cairo | June 2025 – Present
  - **Portfolio Platform Development:** Built an integrated portfolio, event management, HR, and judging system with role-based access, dashboards, analytics, attendance tracking, custom forms, and evaluation workflows to streamline operations and decision-making.

## TECHNICAL SKILLS

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- **Programming Languages:** C/C++, Python, JavaScript, TypeScript, SQL, PHP, Java, Assembly (x86/MIPS), OpenGL, GLSL
- **Frontend & Full-Stack Web:** React.js, Next.js, Redux Toolkit, Tailwind CSS, Three.js, Node.js, Express.js, RESTful APIs, GraphQL, PostgreSQL, MongoDB, Laravel
- **AI & Machine Learning:** PyTorch, TensorFlow, Keras, OpenCV, YOLO, Deep Learning, CNNs, Image Processing, NumPy, Pandas, Scikit-learn
- **Robotics, IoT & Systems:** ESP32, Arduino, Unix/Linux Development, Multithreading, Memory Management, Network Socket Programming, WebSockets
- **DevOps & Core Concepts:** Docker, Git Version Control, CI/CD Pipelines, Object-Oriented Programming, Microservices, System Architecture, Algorithms